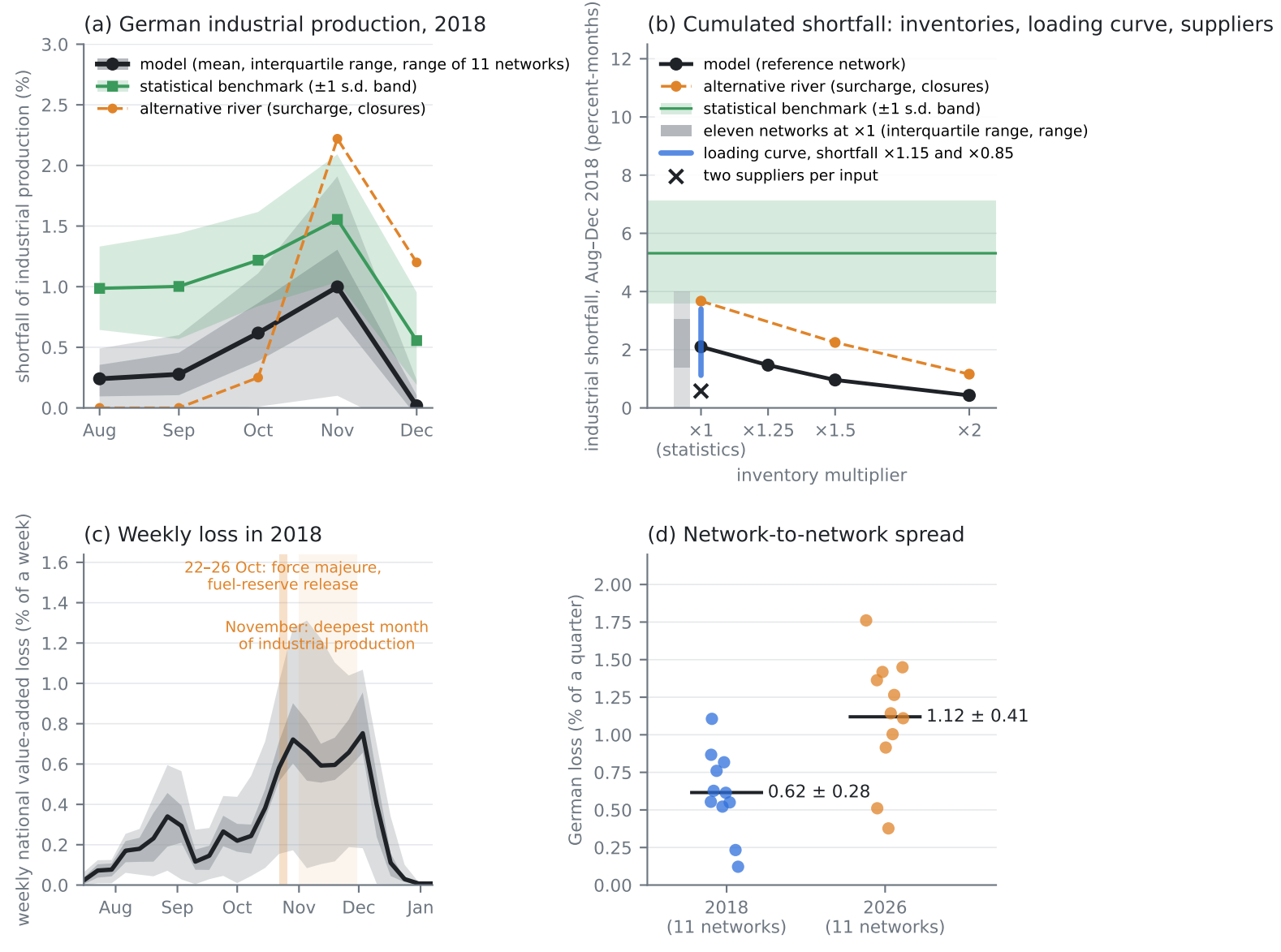
(a) German industrial production, 2018
model (mean, interquartile range, range of 11 networks)
statistical benchmark (±1 s.d. band)
alternative river (surcharge, closures)
shortfall of industrial production (%)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Aug
Sep
Oct
Nov
Dec
(b) Cumulated shortfall: inventories, loading curve, suppliers
model (reference network)
alternative river (surcharge, closures)
statistical benchmark (±1 s.d. band)
eleven networks at ×1 (interquartile range, range)
loading curve, shortfall ×1.15 and ×0.85
two suppliers per input
industrial shortfall, Aug–Dec 2018 (percent-months)
12
10
8
6
4
2
0
×1
(statistics)
×1.25
×1.5
×2
inventory multiplier
(c) Weekly loss in 2018
22–26 Oct: force majeure,
fuel-reserve release
November: deepest month
of industrial production
weekly national value-added loss (% of a week)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Aug
Sep
Oct
Nov
Dec
Jan
(d) Network-to-network spread
German loss (% of a quarter)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
1.12 ± 0.41
0.62 ± 0.28
2018
(11 networks)
2026
(11 networks)